Single subject QC report generated using eddy quad v1.0.3

When using eddy and its QC tools, we ask you to please reference the papers describing the different aspects of the modelling and corrections. The following suggestion for a methods section and list of references has been tailored for you based on your eddy command line.

METHODS
Eddy current-induced distortions and gross subject movement were corrected using the "eddy" tool (Andersson & Sotiropoulos, 2016). The quality of the dataset was assessed using the eddy QC tools (Bastiani et al., 2019). Slices with signal loss caused by subject movement coinciding with the diffusion encoding were detected and replaced by predictions made by a Gaussian Process (Andersson et al., 2016).

REFERENCES
Jesper L.R. Andersson and Stamatios N. Sotiropoulos. 2016. An integrated approach to correction for off-resonance effects and subject movement in diffusion MR imaging. NeuroImage 125:1063-1078

Matteo Bastiani, Michiel Cottaar, Sean P. Fitzgibbon, Sana Suri, Fidel Alfaro-Almagro, Stamatios N. Sotiropoulos, Saad Jbabdi and Jesper L.R. Andersson. 2019. Automated quality control for within and between studies diffusion MRI data using a non-parametric framework for movement and distortion correction. NeuroImage 184:801-812

Jesper L.R. Andersson, Mark S. Graham, Eniko Zsoldos and Stamatios N. Sotiropoulos. 2016. Incorporating outlier detection and replacement into a non-parametric framework for movement and distortion correction of diffusion MR images. NeuroImage 141:556-572

Volume-to-volume motion

.	Average abs. motion (mm)	0.64
.	Average rel. motion (mm)	0.35
.	Average x translation (mm)	0.30
.	Average y translation (mm)	-0.28
.	Average z translation (mm)	0.25
.	Average x rotation (deg)	-0.21
.	Average y rotation (deg)	-0.28
.	Average z rotation (deg)	-0.06

Outliers

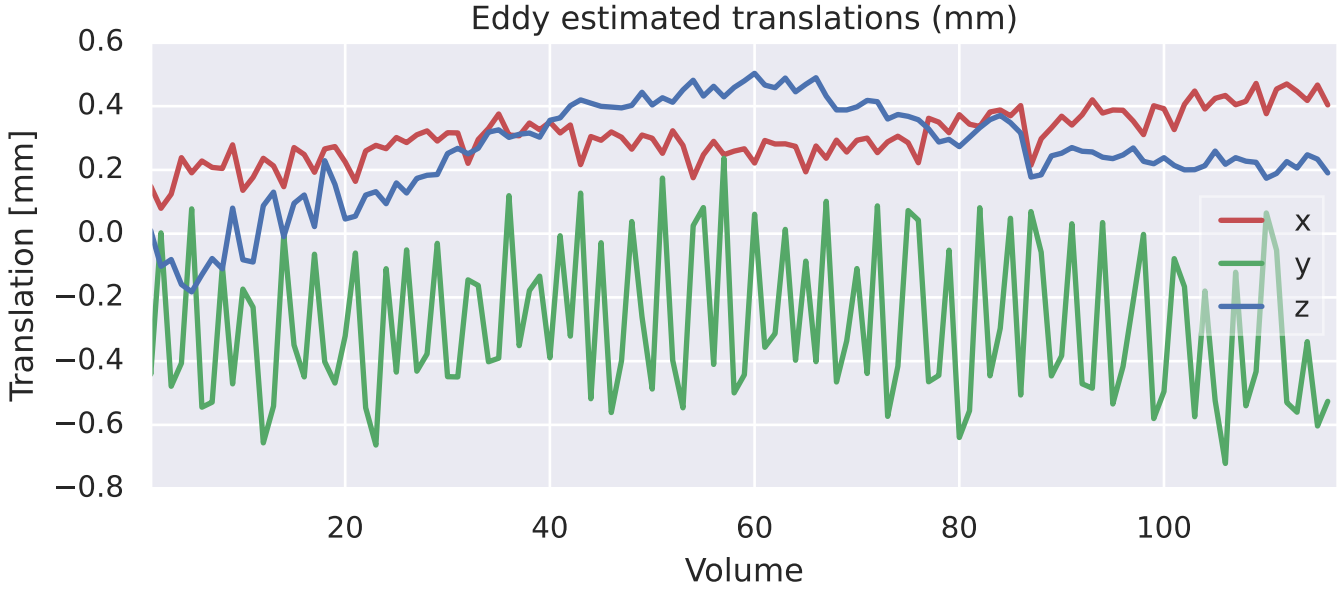
.	Total outliers (%)	0.64
.	Outliers (b=700 s/mm^2)	3.19
.	Outliers (b=1000 s/mm^2)	1.49
.	Outliers (b=2850 s/mm^2)	0.00
.	Outliers (PE dir=[0. -1. 0.])	0.57

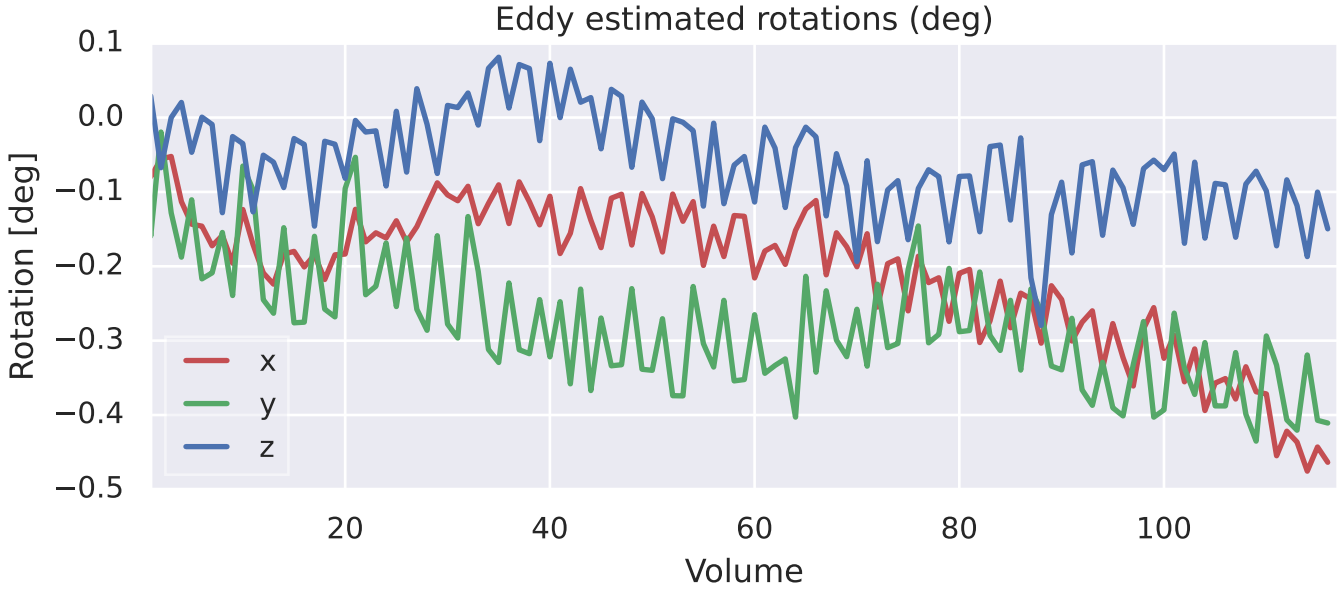
SNR/CNR

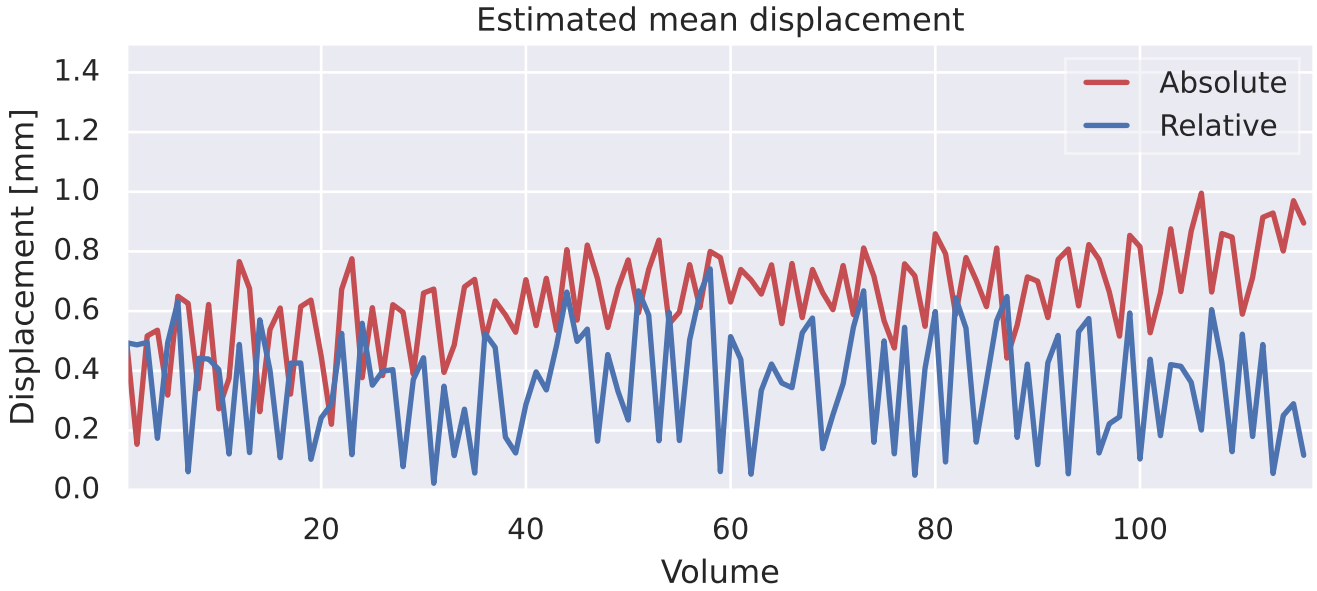
.	Average SNR (b=0 s/mm^2)	18.17
.	Average CNR (b=700 s/mm^2)	0.88
.	Average CNR (b=1000 s/mm^2)	0.78
.	Average CNR (b=2850 s/mm^2)	0.90

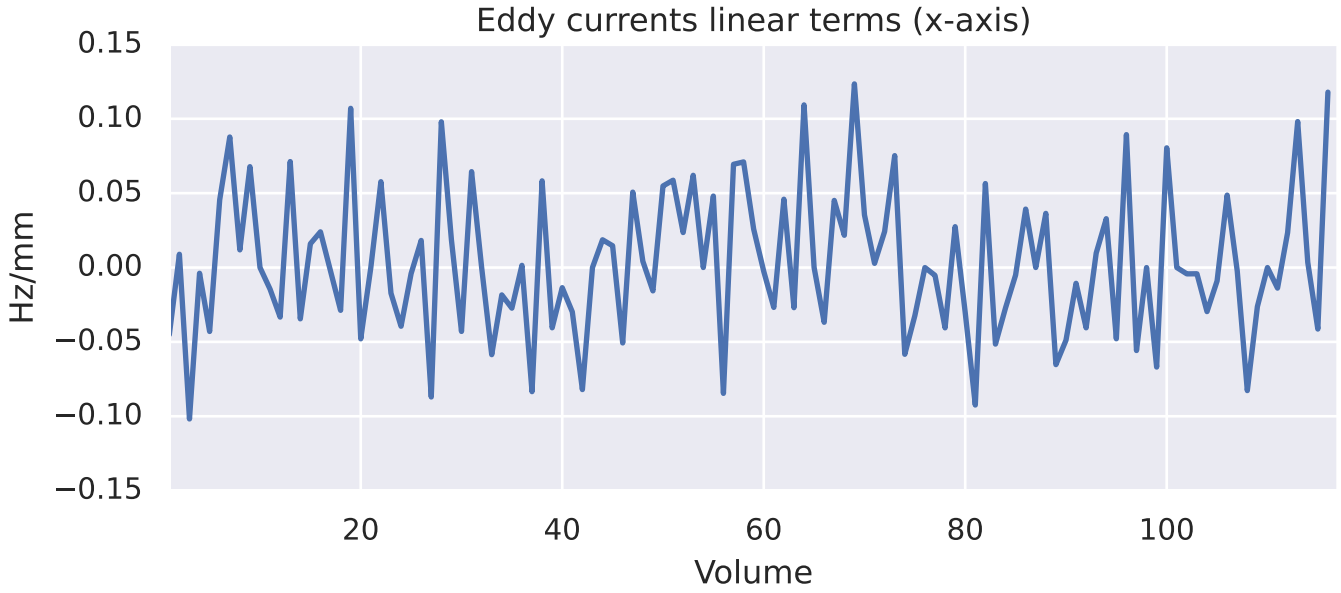
Eddy currents

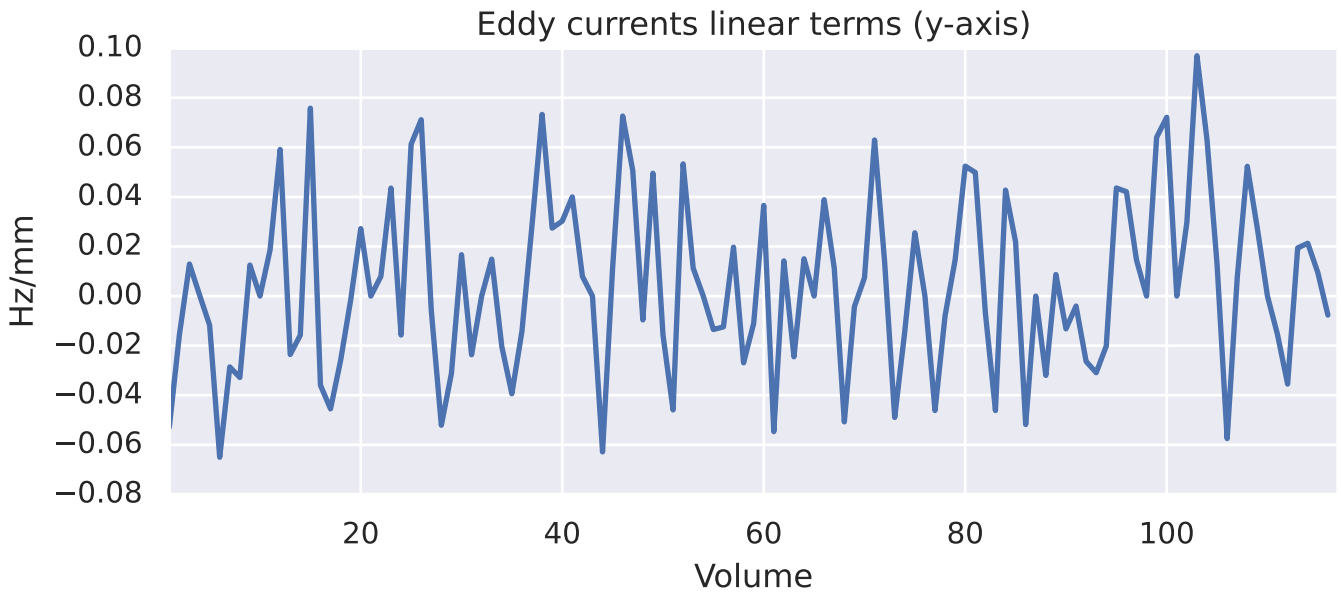
.	Std Dev EC linear term (x)	0.05
.	Std Dev EC linear term (y)	0.04
.	Std Dev EC linear term (z)	0.06

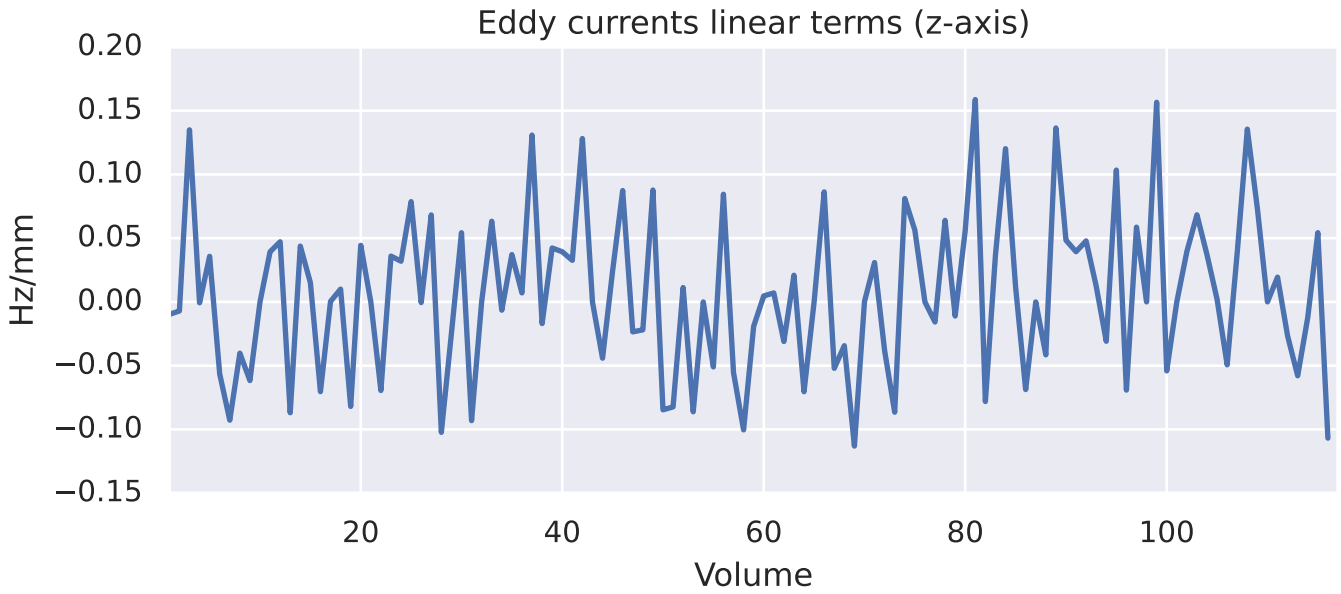

Eddy estimated translations (mm)
Translation [mm]
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
20
40
60
80
100
Volume
x
y
z


Eddy estimated rotations (deg)
Rotation [deg]
0.1
0.0
−0.1
−0.2
−0.3
−0.4
−0.5
20
40
60
80
100
Volume
x
y
z

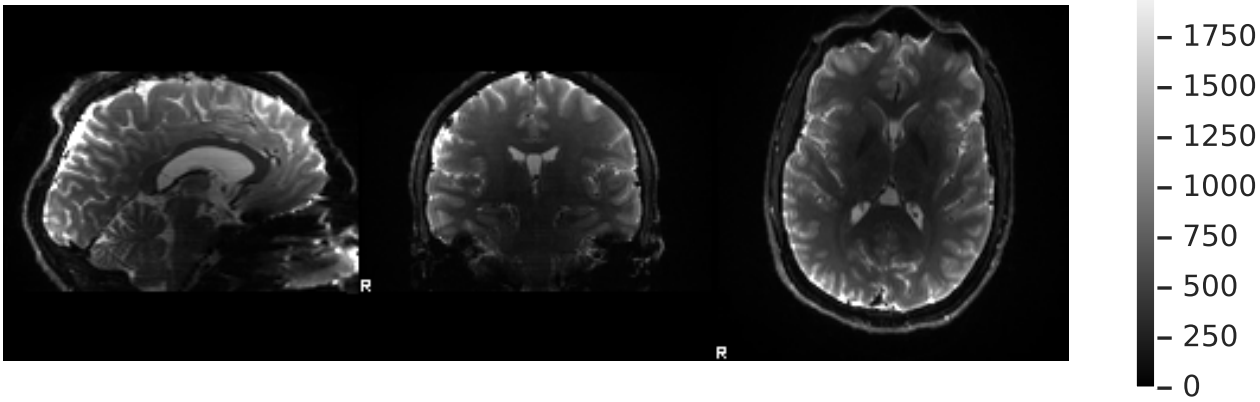

Estimated mean displacement
Displacement [mm]
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
20
40
60
80
100
Volume
Absolute
Relative


Eddy currents linear terms (x-axis)
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Hz/mm
20
40
60
80
100
Volume

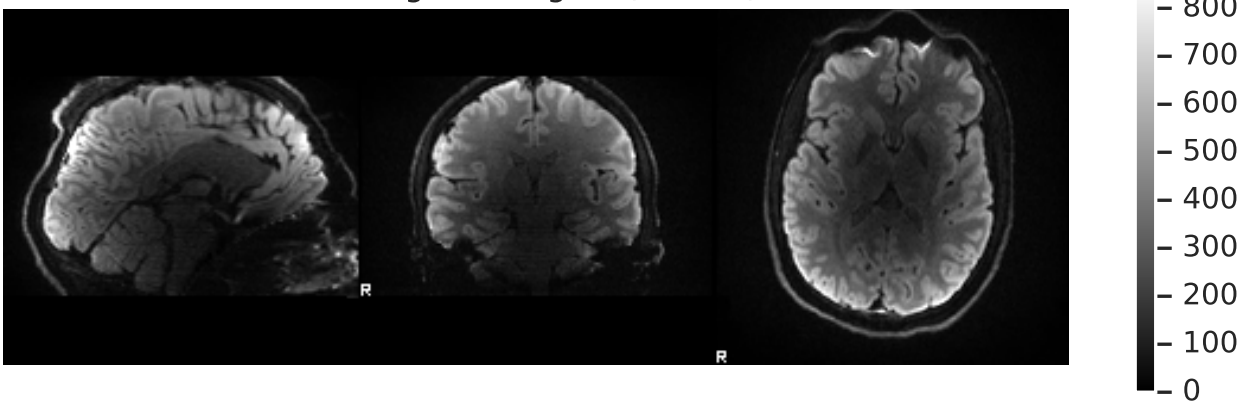

Eddy currents linear terms (y-axis)
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
−0.08
Hz/mm
20
40
60
80
100
Volume


Eddy currents linear terms (z-axis)
0.20
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Hz/mm
20
40
60
80
100
Volume

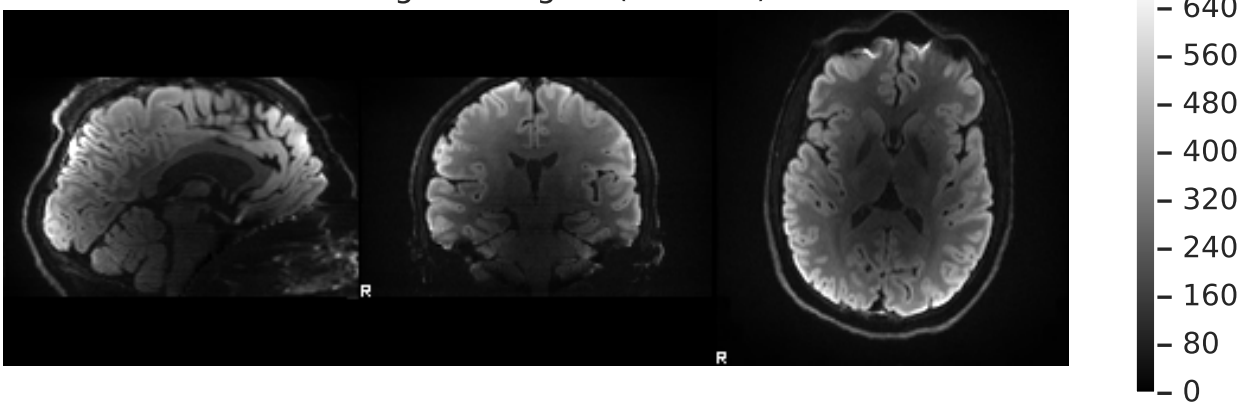
Average DW signal (b=0)



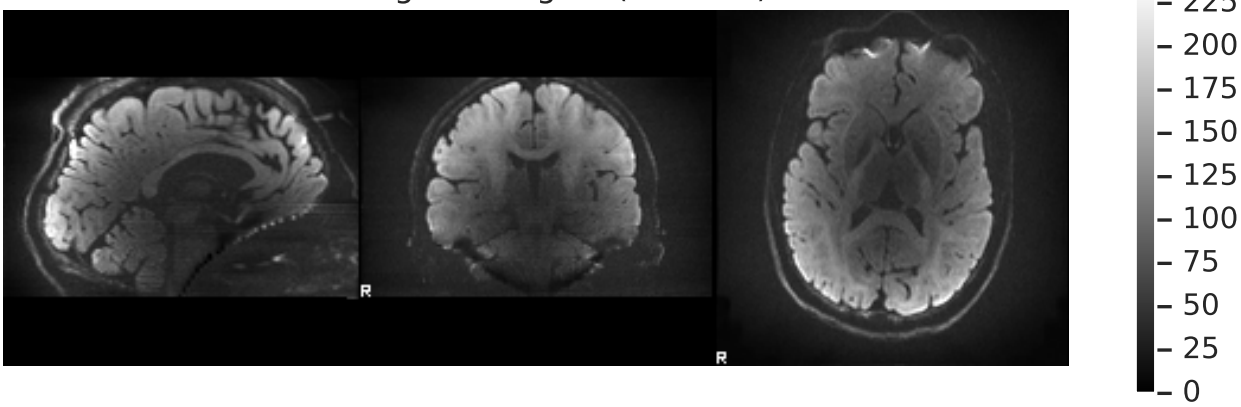
Average DW signal (b=700)

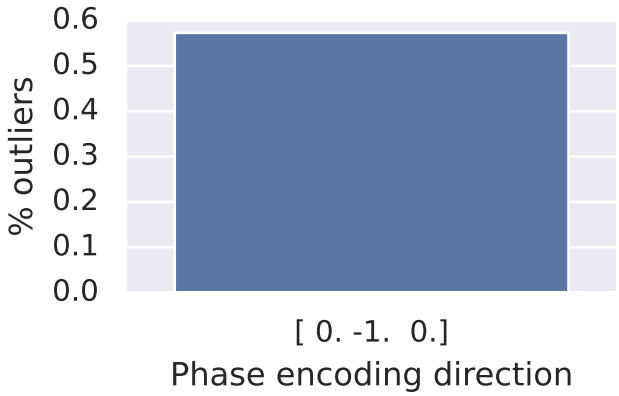


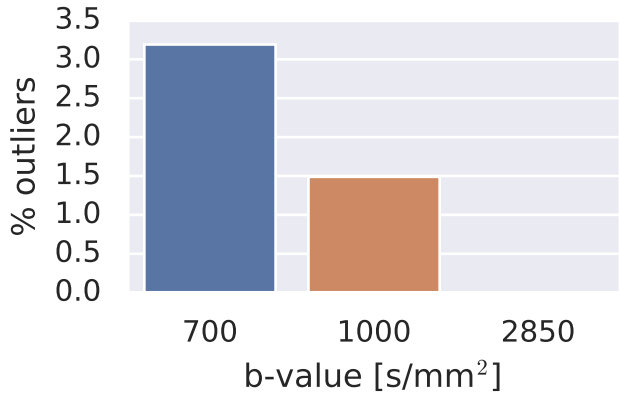
Average DW signal (b=1000)

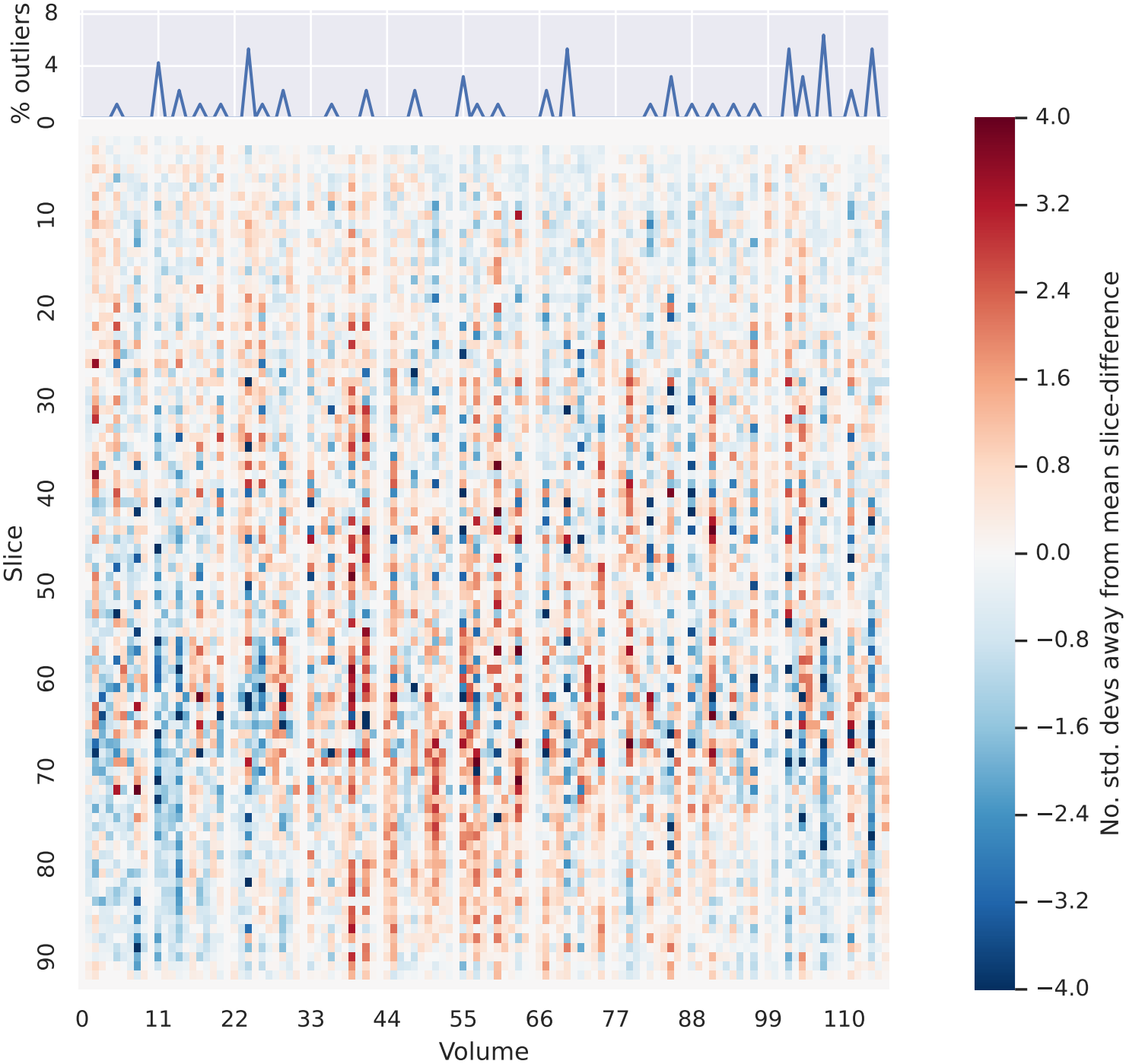


Average DW signal (b=2850)

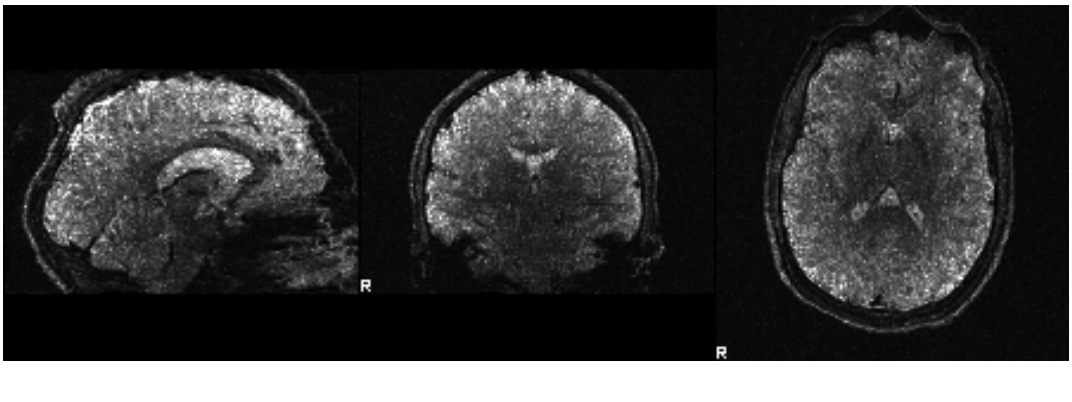



% outliers
0.6
0.5
0.4
0.3
0.2
0.1
0.0
[0. -1. 0.]
Phase encoding direction

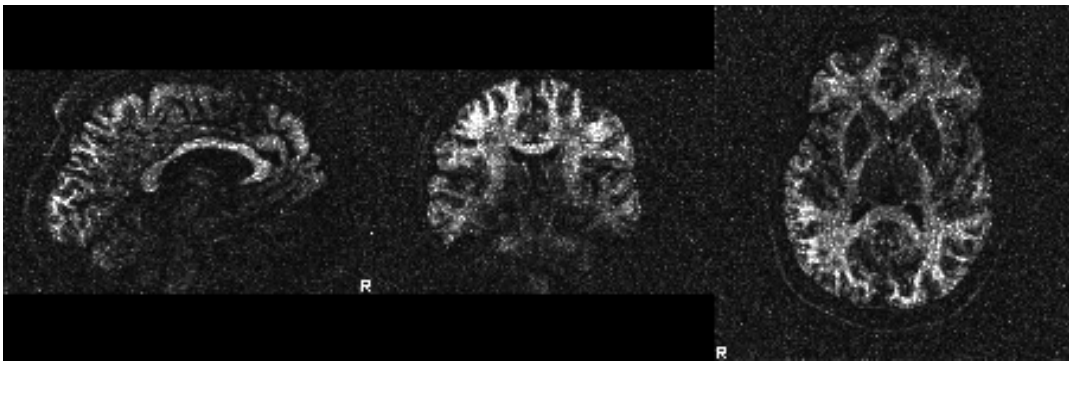

% outliers
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
700
1000
2850
b-value [s/mm^2]


% outliers
8
4
0
Slice
10
20
30
40
50
60
70
80
90
0
11
22
33
44
55
66
77
88
99
110
Volume
4.0
3.2
2.4
1.6
0.8
0.0
−0.8
−1.6
−2.4
−3.2
−4.0
No. std. devs away from mean slice-difference

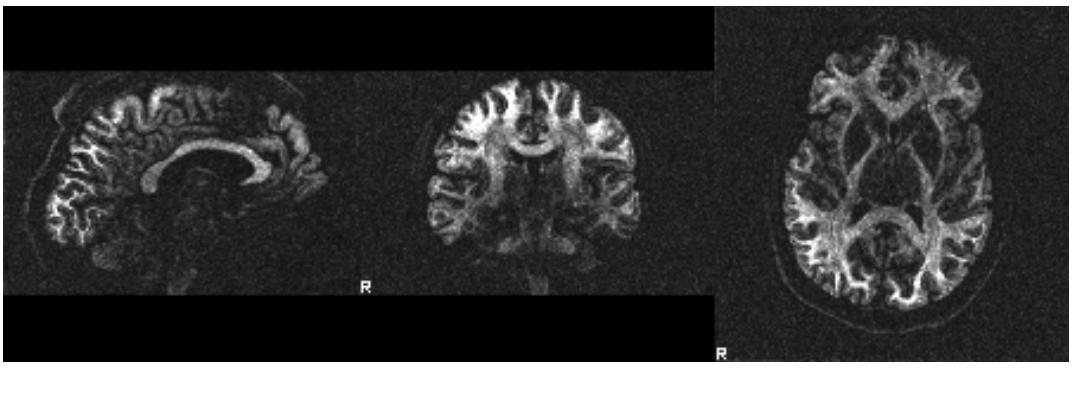
tSNR map (b=0)



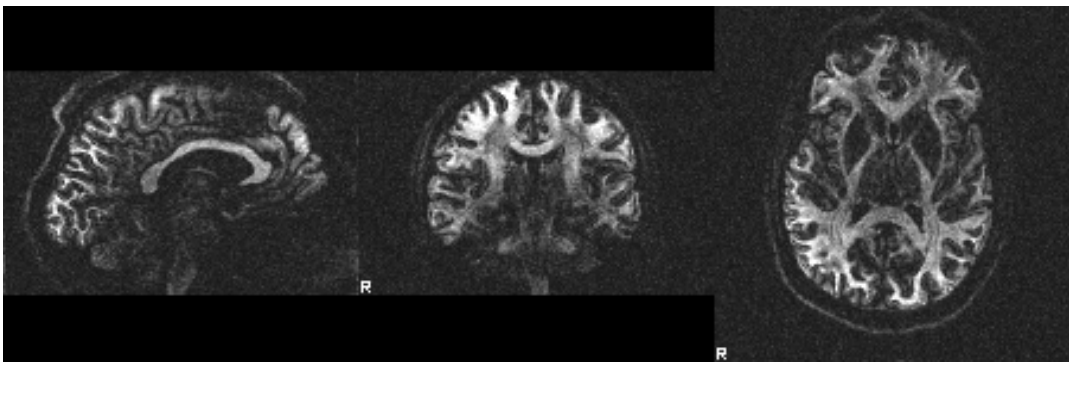
CNR map (b=700)



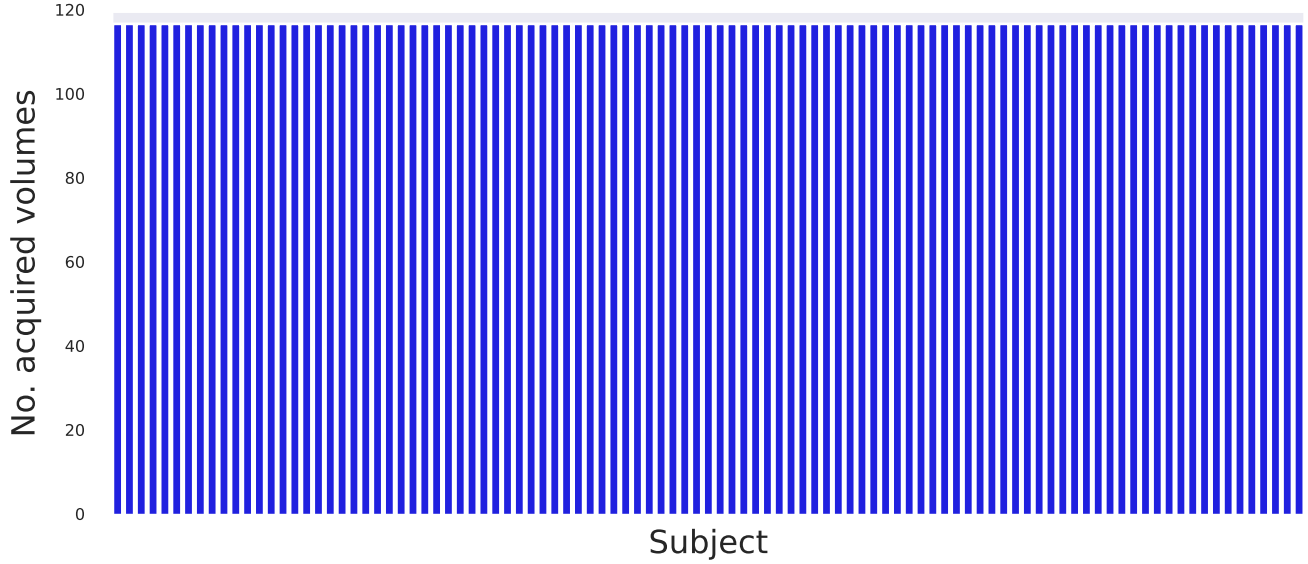
CNR map (b=1000)

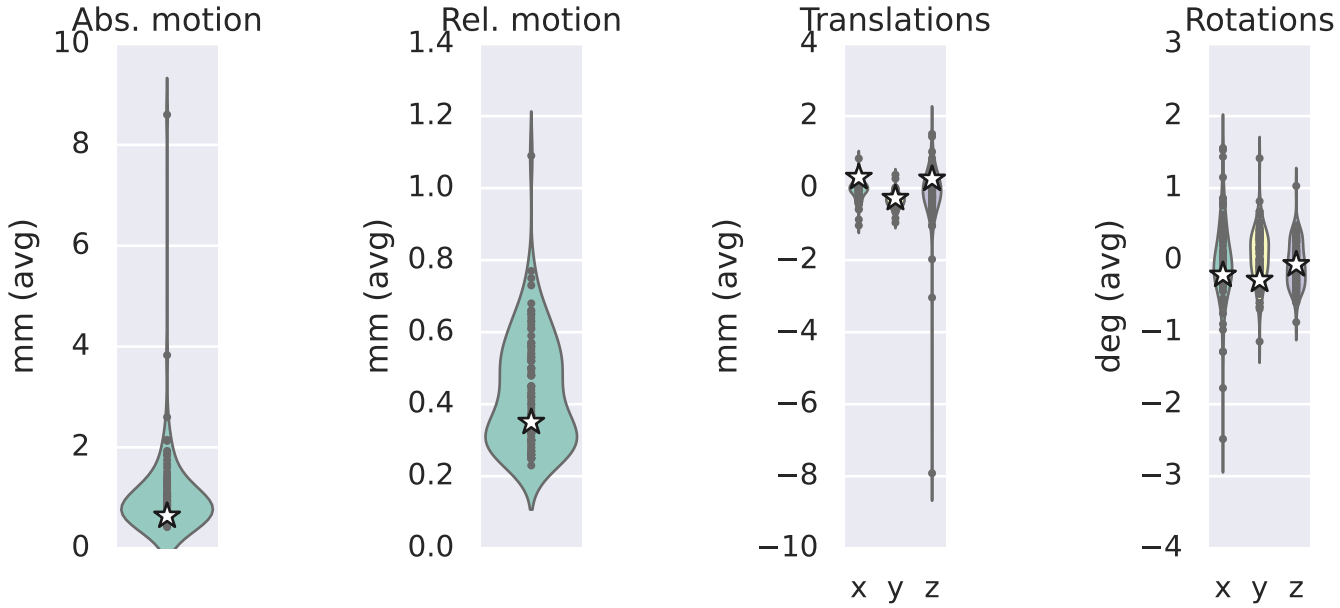


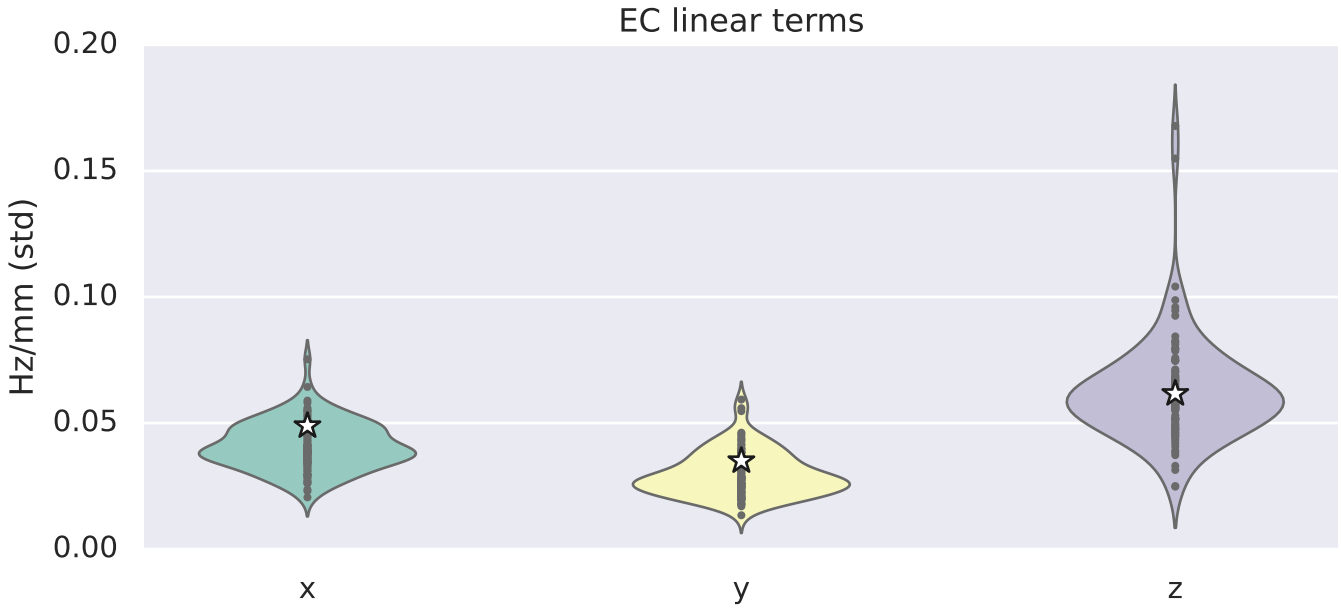
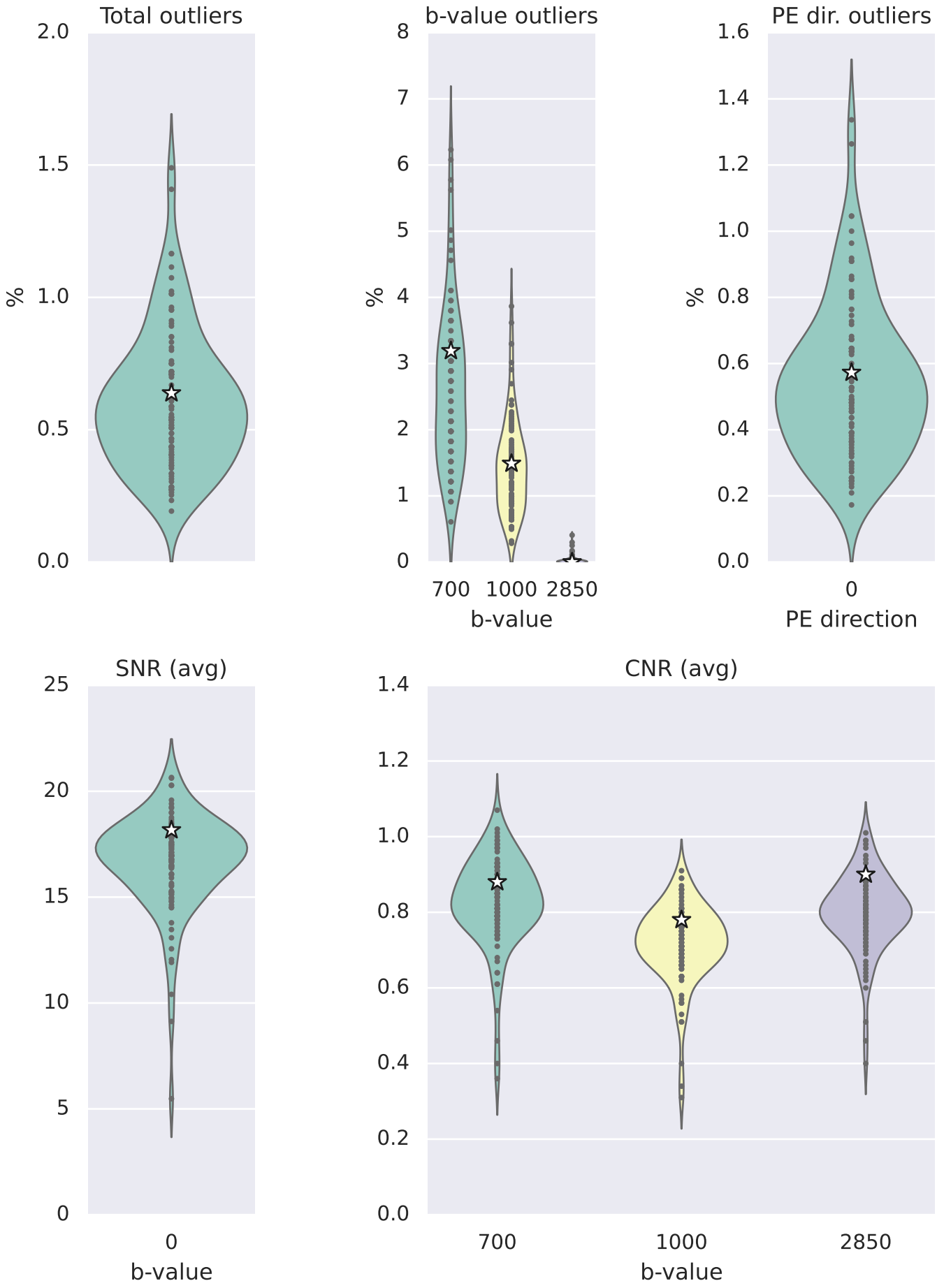
CNR map (b=2850)



SQUAD: Group report







SQUAD: Group report